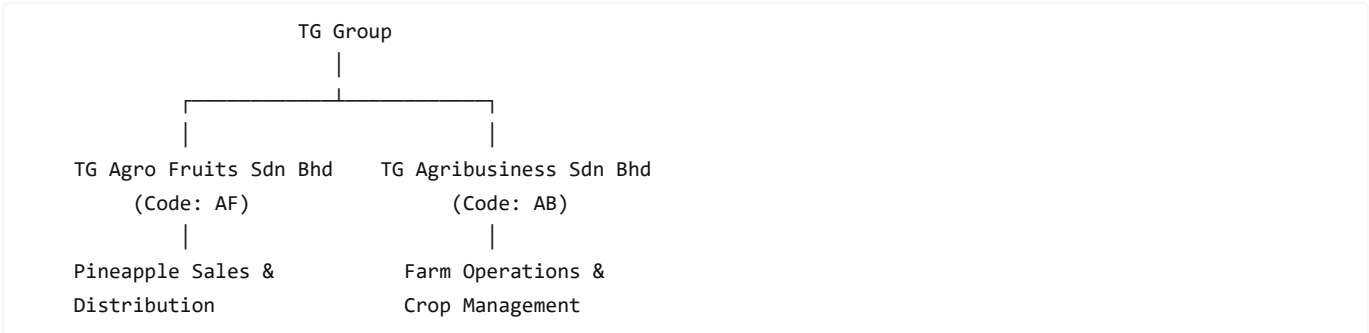


TG FarmHub — Organizational Structure

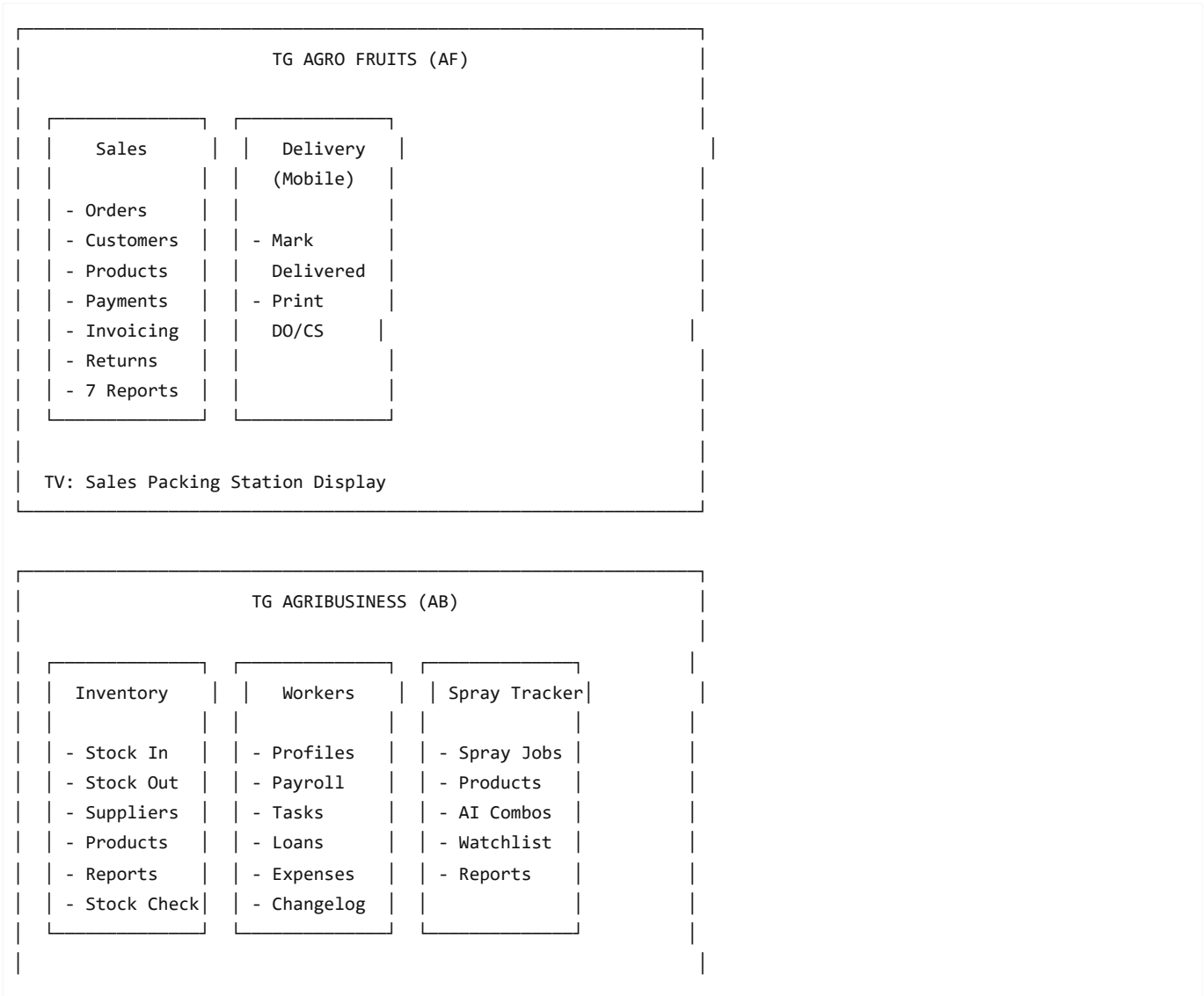
Generated 2026-04-10. Reference doc for planning.

1. Company Structure



Intercompany relationship: TG Agribusiness grows pineapples at Ladang PND. TG Agro Fruits buys from Agribusiness (monthly bulk purchase) and sells to external customers at markup. Intercompany billing is not yet tracked in the system.

2. Module Ownership by Company



Growth Tracker (Read-Only)	Seedlings (Oil Palm)
- Block Growth	- Batches
- Harvest Planning	- Bookings
- Status Dashboard	- Collections
	- L3.1
	- Customers
	- MPOB Rpts

TV: Growth Tracker Display, Spray Monitoring Display

SHARED

Farm Config	TV Display (Links)	Hub
- Blocks		- Login
- Crops	- Growth TV	- Users
- Varieties	- Weather	- Company Switcher
- Statuses	- Sales TV	- Overview
- Planting Cycles	- Spray TV	

3. User Roles & Access

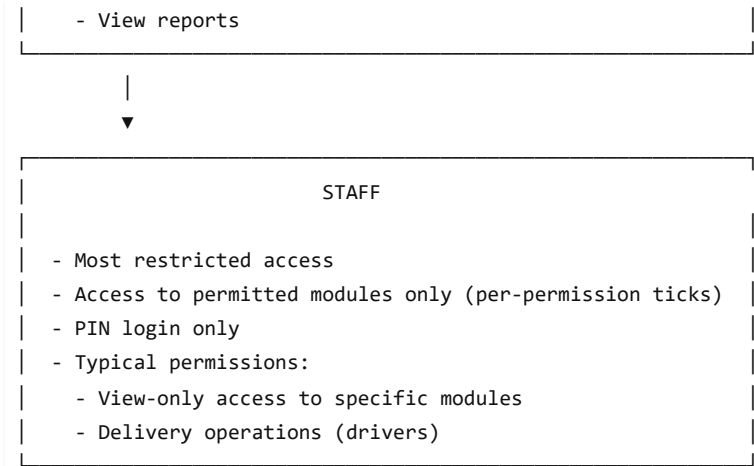
ADMIN

- Full access to all modules in both companies
- User management (create/edit/delete users)
- Farm Configuration (blocks, crops, varieties)
- Invoice approval
- Google OAuth login (yapwaylon@gmail.com)
- PIN login (fallback)
- TV Display configuration



SUPERVISOR

- Access to permitted modules only (per-permission ticks)
- Can perform operations within granted permissions
- PIN login only
- Typical permissions:
 - Log spray applications
 - Record stock in/out
 - Create/edit orders



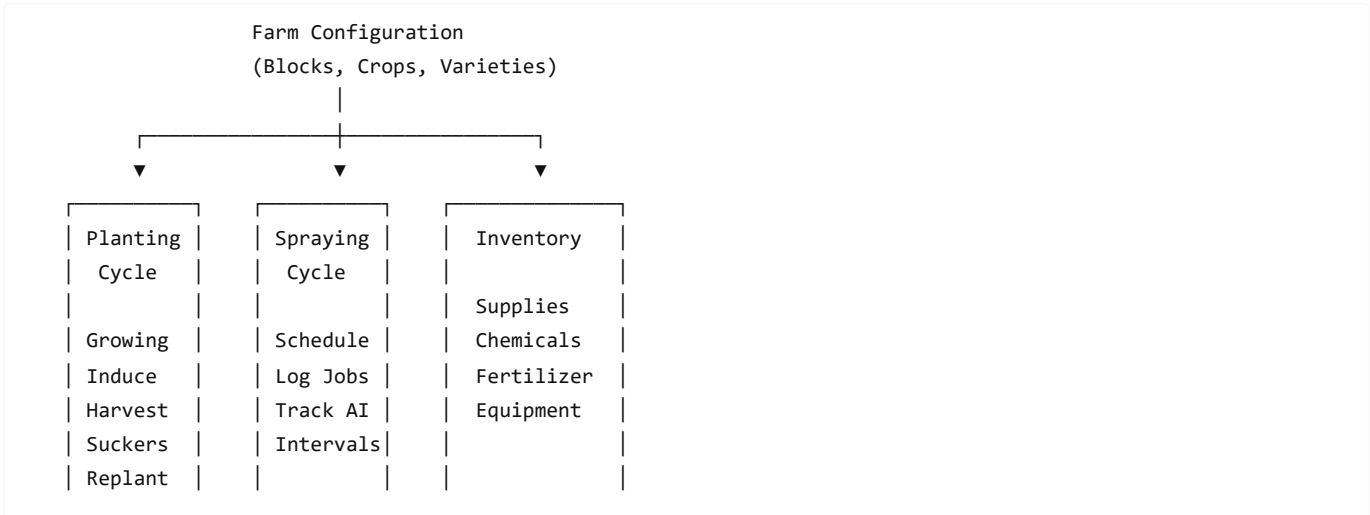
Permission Matrix

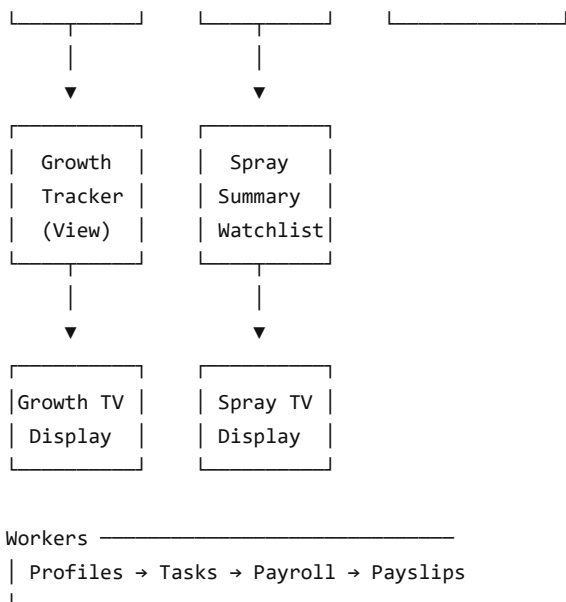
Each module has granular permissions that can be toggled per user:

Module	Permissions
Sales	View, Create/Edit Orders, Payments, Delivery, Manage Customers, Export
Inventory	View, Stock In, Stock Out, Stock Check, Edit Products, Manage Suppliers, Edit Transactions, Delete Transactions, Export
Workers	View, Edit Workers, Payroll, Summary/Reports
Spray Tracker	View, Log Sprays, Manage Products
Growth Tracker	View, Manage Growth (induction/harvest dates)
Farm Config	Manage Farm (blocks/crops/varieties)
TV Display	View (access TV Display links)
Seedlings	View, Manage Batches, Manage Bookings, Record Collections, Manage Customers, Manage Suppliers, View Reports

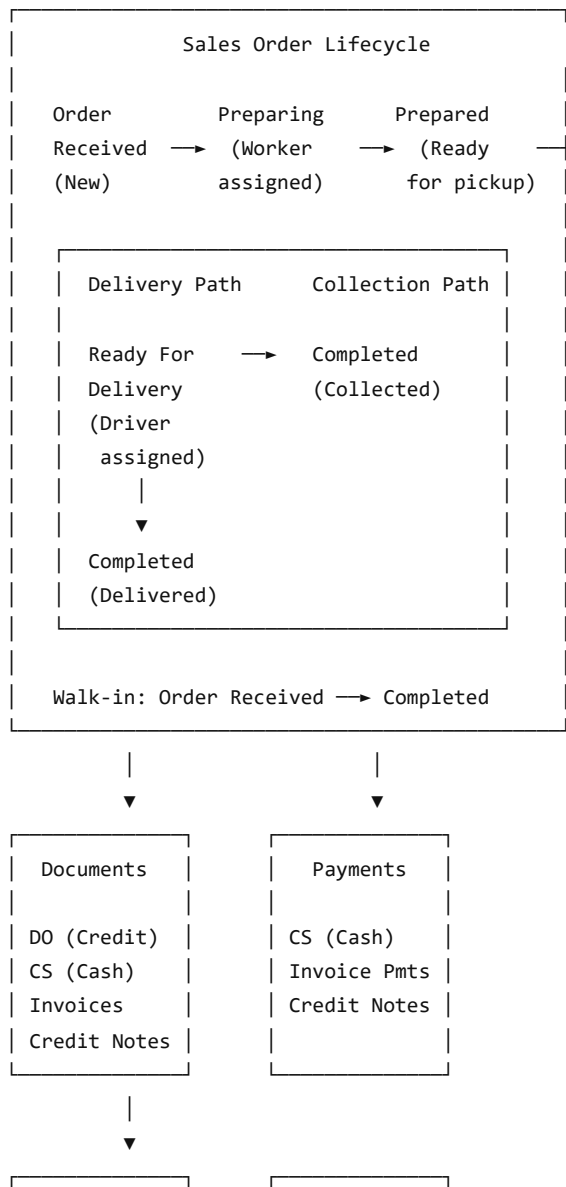
4. Operational Workflow

Agribusiness — Farm Operations



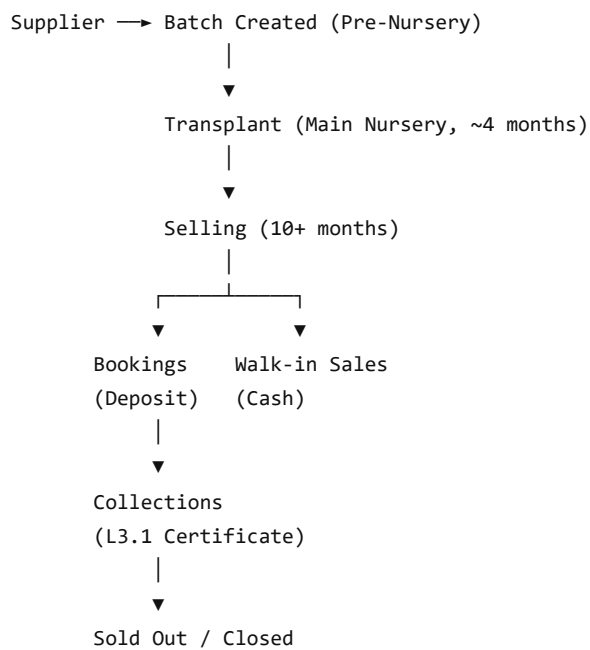


Agro Fruits — Sales Operations

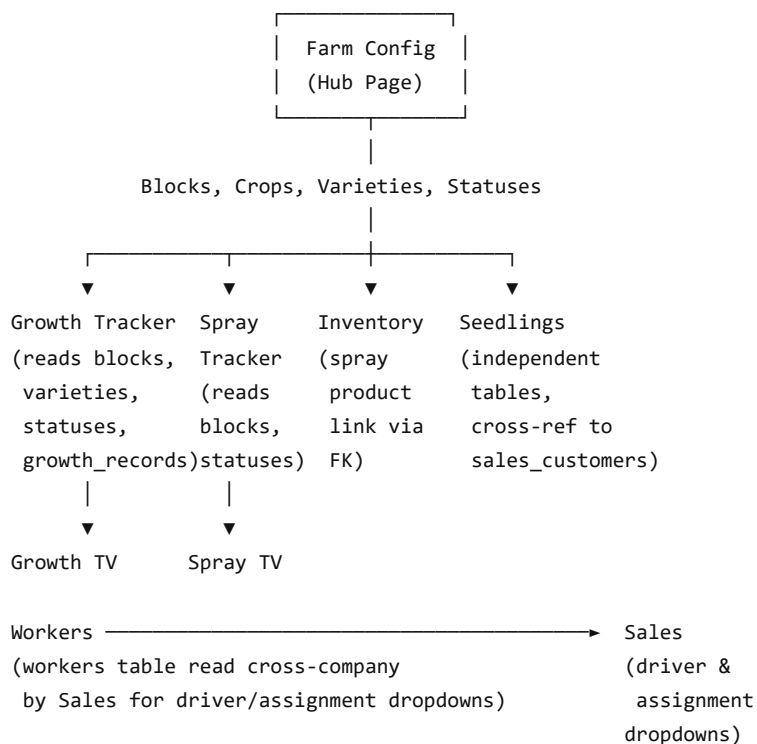


Packing	Delivery
Station TV	(Mobile)
4-column	Driver app
live grid	Mark done

Agribusiness — Seedlings Operations



5. Data Flow Between Modules



Cross-Module Data Dependencies

Source	Consumer	What's Shared
Farm Config → Growth Tracker	Blocks, varieties, statuses, growth_records	Block lifecycle data
Farm Config → Spray Tracker	Blocks, statuses, varieties	Block info for spray logging
Inventory → Spray Tracker	pnd_products.inventory_product_id FK	Product stock levels
Workers → Sales, Delivery	workers table (no company filter)	Driver names, worker assignments
Sales Customers → Seedlings	sales_customers (by phone match)	Cross-company customer link
Farm Config → Growth TV	growth_records, pnd_blocks, crop_varieties	TV display data
Hub → Spray TV	pnd_tv_config (watched AIs)	TV display configuration

6. Physical Locations

Location	Address	Used For
Office	Lot 1609, Kpg. Riam Jaya, 98000 Miri, Sarawak	Business address, booking slips
Farm / MPOB License	Lot 174, Block 9, Lambir Land District, 98000 Miri, Sarawak	L3.1 certificates, farm operations
MPOB License No.	522231011000	Seedlings compliance

7. External Integrations

System	Integration	Status
Supabase	Database, Auth, Storage	Active
Google OAuth	Admin login	Active
WhatsApp	Message sharing (orders, spray jobs, documents)	Active (via Web Share API)
Netlify	Hosting & deploy	Active
MPOB	Monthly seedling reports (manual/printable)	Active
QuickBooks	Removed (replaced by in-house invoicing)	N/A

8. TV Display Network

Hub Page (TV Display Tab)	
├─ Growth Tracker TV → nanasgrowth.tgfarmhub.com	
├─ (Separate repo) Auto-rotate by crop status	
├─ Weather TV → weather.tgfarmhub.com	
├─ (Separate deploy) Weather monitoring	
├─ Sales Packing TV → tgfarmhub.com/display-sales.html	
├─ (Same repo) 4-column order grid	

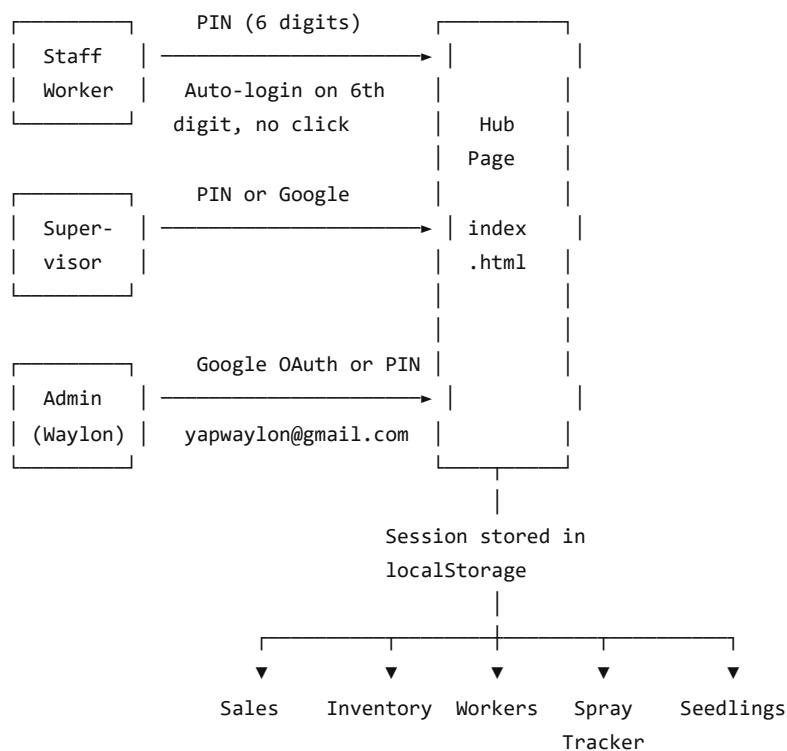
└ Spray Tracker TV → tgfarmhub.com/display-spray.html
(Same repo) Per-AI spray monitoring

All TVs: Password-gated, read-only, auto-refresh

9. Document Types Generated

Document	Format	Module	Delivery
Delivery Order (DO)	80mm receipt + A4	Sales	Print / WhatsApp PNG
Cash Sales (CS)	80mm receipt + A4	Sales	Print / WhatsApp PNG
Invoice	A4 (multi-page with DO summary)	Sales	Print / WhatsApp PNG
Credit Note	A4	Sales	Print / WhatsApp PNG
Statement of Account	A4	Sales	Print / WhatsApp PNG
Debit Note	Numbered reference	Sales	—
Booking Slip	Printable	Seedlings	Print / WhatsApp
Payslip	A5 (2 per A4 landscape)	Workers	Print
MPOB Monthly Report	Printable table	Seedlings	Print / CSV
Batch Summary	Printable table	Seedlings	Print / CSV
Spray Reports (8 types)	Printable table	Spray Tracker	Print

10. Authentication Flow Summary



Security:

- |— Multi-device: login elsewhere logs out previous (polled 30s)
- |— Inactivity timeout: 60 minutes
- |— RLS on all tables (anon + authenticated roles)
- |— Permissions: per-module granular toggles